

6 Fig. 6.1 shows a circuit that rectifies an alternating input voltage V_{IN} and produces an output voltage V_{OUT} across a resistor R .

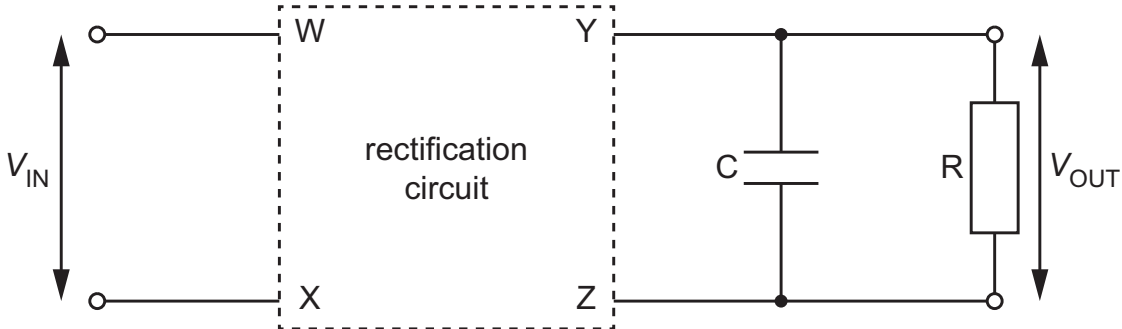


Fig. 6.1

The four terminals of the rectification circuit are labelled W, X, Y and Z.
A capacitor C is connected in parallel with resistor R.

(a) (i) State what is meant by rectification.

.....
 [1]

(ii) State the purpose of capacitor C.

.....
 [1]

(b) Fig. 6.2 shows the variations with time t of the potential differences (p.d.s) V_{IN} and V_{OUT} .

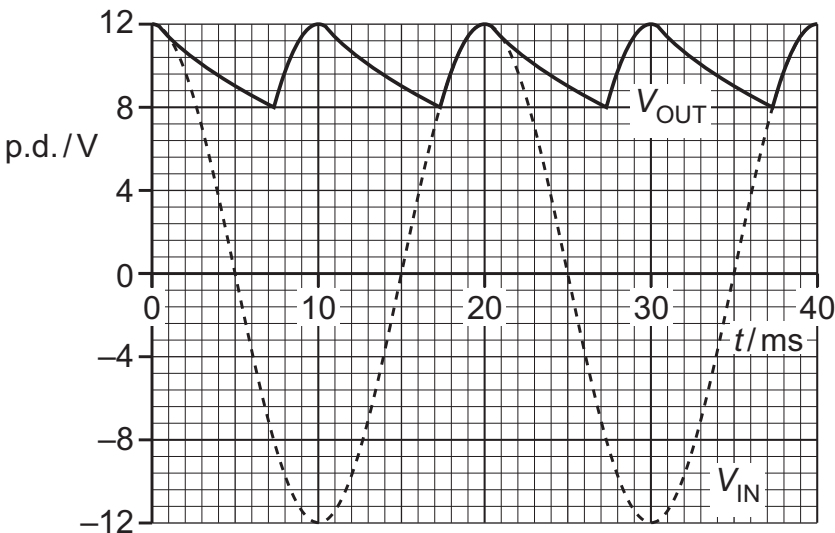


Fig. 6.2

(i) The variation of V_{IN} with t can be represented by

$$V_{\text{IN}} = A \cos Bt$$

where A and B are constants.

Determine the values of A and B . Give a unit with your answer for A .

$A =$ unit

$B =$ rad s^{-1}
 [2]

(ii) Determine the type of rectification produced by the circuit in Fig. 6.1.

..... [1]

(iii) On Fig. 6.3, draw the circuit diagram for the components inside the rectification circuit.

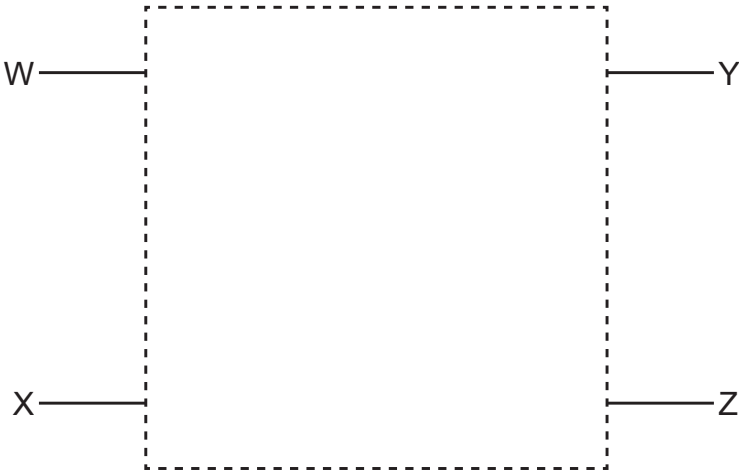


Fig. 6.3

[2]

(iv) Determine a value for the time constant for the discharge of the capacitor C through the resistor R in Fig. 6.1.

time constant = s [3]

(c) The capacitor C has a capacitance of $570\,\mu\text{F}$.

Use your answer in (b)(iv) to determine the resistance of resistor R.

resistance = Ω [2]